Activity

Select a leafy twig of fully grown plant in your school campus. Cover the twig with a transparent polythene bag and tie the mouth of the bag at the base of the twig. Observe the changes after two hours and discuss with your teacher

11.6.1 Types of Transpiration

Transpiration is of following three types:

1. Stomatal transpiration

Stomata are microscopic structures present in high number on the lower epidermis of leaves. This is the most dominant form of transpiration and being responsible for most of the water loss (90 - 95%) in plants.

2. Lenticular transpiration

In stems of woody plants and trees, the epidermis is replaced by periderm because of secondary growth. In order to provide gaseous exchange between the living cells and outer atmosphere, some pores which looks like lens-shaped raised spots are present on the surface of the stem called **Lenticels**. The loss of water from lenticels is very insignificant as it amounts to only 0.1% of the total.

3. Cuticular transpiration

The cuticle is a waxy or resinous layer of **cutin**, a fatty substance covering the epidermis of leaves and other plant parts. Loss of water through cuticle is relatively small and it is only about 5 to 10 % of the total transpiration. The thickness of cuticle increases in xerophytes and transpiration is very much reduced or totally absent.

11.6.2 Structure of Stomata

The epidermis of leaves and green stems possess many small pores called **stomata**. The length and breadth of stomata is about 10-40µ and 3-10µ respectively. Mature leaves contain between 50 and 500 stomata per square mm. Stomata are made up of two **guard cells**, special semi-lunar or kidney-shaped living epidermal cells in the epidermis. Guard cells are attached to surrounding epidermal cells known as **subsidiary cell**s or **accessory cells**. The guard cells are joined together at each end but they are free to separate to form a pore between them. The inner wall of the guard cell is thicker than the outer wall (Figure 11.14). The stoma opens to the interior into a cavity called **sub-stomatal cavity** which remains connected with the intercellular spaces.

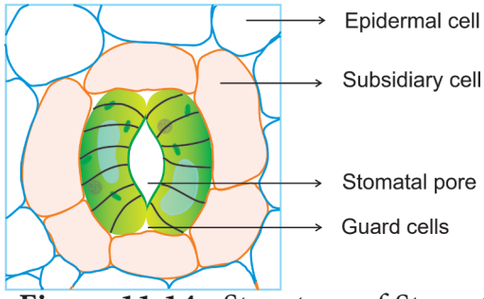


Figure 11.14: Structure of Stomata

11.6.3 Mechanism of Stomatal Movement

Stomatal movements are regulated by the change of turgor pressure in guard cells. When water enters the guard cell, it swells and its unevenly thickened walls stretch up resulting in the opening of stomata. This is due to concave non-elastic nature of inner wall pulled away from each other and stretching of the convex elastic natured outer wall of guard cell.

Different theories have been proposed regarding opening and closing of stomata. The important theories of stomatal movement are as follows,

1. *Theory of Photosynthesis in guard cells*
2. *Starch – Sugar interconversion theory*
3. *Active potassium transport ion concept*

1. Theory of Photosynthesis in guard cells

Von Mohl (1856) observed that stomata open in light and close in the night. According to him, chloroplasts present in the guard cells photosynthesize in the presence of light resulting in the production of carbohydrate (Sugar) which increases osmotic pressure in guard cells. It leads to the entry of water from other cell and stomatal aperture opens. The above process *vice versa* in night leads to closure of stomata.

Demerits

1. Chloroplast of guard cells is poorly developed and incapable of performing photosynthesis.

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